



Mustain Billah Taj

Backend & AI Systems Engineer

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PROFILE

Software Engineering student focused on **backend development** and **agentic, LLM-powered applications**. I build REST APIs and real-time services in Python and FastAPI, with clean service boundaries and background workers that keep the system responsive. On the AI side, I build retrieval-augmented and agentic systems with LangChain and LangGraph — routing, tool use, vector search.

TECHNICAL SKILLS

Languages	Python, Java, C++, TypeScript	Backend	FastAPI, REST APIs, WebSocket, WebRTC, SSE
AI / LLM	LangChain, LangGraph, Groq, RAG, pgvector	Databases	PostgreSQL, MySQL, MongoDB, Redis
ML / DL	PyTorch, pandas, ANN, CNN, RNN, LSTM, Regression, Random Forest, XGBoost	Frontend	Next.js, React, Tailwind CSS
Core CS	Data Structures, Algorithms, OOP, System Design	Tools	Docker, Git, Stripe, Alembic, arq
Auth	JWT, Argon2 / bcrypt, role-based access		

SELECTED PROJECTS

Health Hive

live demo · github.com/Mustain98/Health-Hive

An AI-based meal planner and health guide with real consultants in the loop.

FastAPI LangChain v1 LangGraph Groq PostgreSQL + pgvector SSE Agora Next.js 16

OVERVIEW

An AI-powered meal planner and health guide where users build personalized plans through conversation, and verified consultants step in for guidance, follow-ups, and live video calls.

WHAT I BUILT

- AI-driven health planning through chat.** Users create their plan by talking to an AI chatbot that works from their health data — setting milestones, daily goals, and daily nutrition targets tailored to them.
- Preferences in plain language.** Users describe their food preferences, dietary restrictions, and health conditions in plain text, and the system factors all of it into the plans it generates.
- LLM-powered meal planning.** The LLM derives per-meal nutrition constraints from the user’s targets, vector retrieval finds matching meal candidates, and the plan is assembled to hit the required macros while respecting every dietary restriction.
- Consultant interaction and follow-up.** Users connect with verified nutritionists and doctors, exchange messages, and continue through structured follow-up threads after their sessions.
- Real-time video consultations.** High-touch guidance happens over live video calls, booked directly with a consultant through the platform.

Stream

github.com/Mustain98/stream

A pay-per-view live streaming platform with real-time video and integrated payments.

FastAPI aiortc WebRTC Stripe WebSocket PostgreSQL Docker Next.js 15

OVERVIEW

A pay-per-view platform where creators broadcast live to viewers over WebRTC, offer free previews, and get paid through an end-to-end Stripe checkout and payout flow.

WHAT I BUILT

- **Real-time live streaming over WebRTC.** Broadcasters stream live audio and video to many viewers at once, with room lifecycle, signaling, and preview cutoffs handled in real time.
- **Pay-per-view access with free previews.** Viewers watch a free preview, then unlock full access through checkout — with the preview limit enforced server-side so it can't be bypassed.
- **End-to-end Stripe payments.** Checkout, webhook confirmation, platform fees, creator payouts, and automatic refunds when a stream ends early — with idempotent handling so payments are never processed twice.
- **Clean separation of payments and media.** Business logic and the streaming service are decoupled, linked only by short-lived access tickets, so credentials and payments never touch the media layer.
- **Live chat, moderation, and earnings.** Real-time chat and moderation run alongside the stream, and creators see confirmed earnings update live as payments come in.

J_Buddy

github.com/Mustain98/J_buddy

A job-hunting assistant that reads your CV and does the searching for you.

FastAPI LangChain Groq pgvector Redis + arq PostgreSQL Next.js

THE PROBLEM

Job hunting means guessing search terms, reading hundreds of postings, and honestly judging your own fit for each. I wanted to invert it: the CV does the searching. The system writes its own queries from what is in it, pulls listings in the background, and ranks them against real skills — and an assistant that cheerfully tells you you are qualified for anything is useless, so this one only answers from what the CV actually says.

WHAT I BUILT

- **CV → structured profile.** Upload a PDF or Word file and an LLM decomposes it into skills, experience, projects, education and certifications. No forms.
 - **Self-generating search queries.** It derives its own job queries (“Python backend developer”, “ML intern”) from the CV, learns from searches you run yourself, and reconsiders the whole query set once you have run enough of them.
 - **A hybrid fit score out of 100.** Half lexical — how many of your actual skills and past roles appear in the posting — and half semantic, comparing the whole posting against your profile to catch matches worded differently. Every score ships with a one-sentence justification.
 - **Chat that stays honest.** Grounded in your CV and the job data, able to search live listings mid-conversation and recall jobs discussed earlier, and instructed never to claim skills the CV does not back up.
 - **Preparation, not just applications.** Gap analysis, phased learning roadmaps that skip what you already know with tickable goals, and cover letters written from real experience — “make it shorter, mention Docker” edits the same letter rather than generating a new one.
 - **Nothing slow runs in the request path.** Listing fetches, ranked-pool rebuilds and nightly cleanup all go to an arq worker on Redis, which is what keeps the feed instant. Listings expire after a week; anything on your application board survives even after the original posting is gone.
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